

Calc AB: Integrals Review

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DIRECTIONS

Evaluate each integral using u-substitution, trig identities, or the FTC as needed. For #17-20, integrate or evaluate at the given point. For #21-23, factor and simplify before evaluating the limit.

1 Evaluate:

$$\int_2^4 \frac{15x-3}{5x} dx$$

Answer: _____

2 Evaluate:

$$\int_0^1 x^6(4x^2 + x - 5) dx$$

Answer: _____

3 Evaluate:

$$\int 4\sin^2 x dx$$

Answer: _____

4 Evaluate:

$$\int_3^4 \frac{3x}{x^2-2} dx$$

Answer: _____

5 Evaluate:

$$\int \frac{x^2}{1-x} dx$$

Answer: _____

6 Evaluate:

$$\int \frac{7}{e^{5x}} dx$$

Answer: _____

7 Evaluate:

$$\int \cos^3(7x) dx$$

Answer: _____

8 Evaluate:

$$\int \cos^5 x \sin x dx$$

Answer: _____

9 Evaluate:

$$\int_{\ln 2}^{\ln 9} 4e^x dx$$

Answer: _____

10 Evaluate:

$$\int_4^{25} \frac{e^{\sqrt{x}}}{\sqrt{x}} dx$$

Answer: _____

11 Evaluate:

$$\int_{1/\sqrt{3}}^1 \frac{dx}{1+x^2}$$

Answer: _____

12 Evaluate:

$$\int \cos(3-2x) dx$$

Answer: _____

13 Evaluate:

$$\int \frac{\ln^2 x}{x} dx$$

Answer: _____

14 Evaluate:

$$\int \frac{3x}{(x^2+1)^4} dx$$

Answer: _____

15 Evaluate:

$$\int \frac{3t-t^2}{2t} dt$$

Answer: _____

16 Evaluate:

$$\int \cos^3 2u \sin 2u du$$

Answer: _____

17 Find the antiderivative of:

$$\frac{2\cos x}{\sin^2 x}$$

Answer: _____

18 Given $f'(x)$ and $f(1)=2$, find $f(x)$:

$$f'(x) = \frac{1}{2}x^2 + \frac{3}{4}x$$

Answer: _____

19 Given $f'(x)$ and $f(1)=6$, find $f(2)$:

$$f'(x) = 5x^4 - 2x$$

Answer: _____

20 Simplify:

$$e^{2\ln x^3}$$

Answer: _____

21 Evaluate the limit:

$$\lim_{x \rightarrow 4} \frac{x^2 - 4}{x^2 - 16}$$

Answer: _____

22 Evaluate the limit:

$$\lim_{x \rightarrow 6} \frac{x-6}{x^2-36}$$

Answer: _____

23 Evaluate the limit:

$$\lim_{x \rightarrow 4} \frac{x^3 - 64}{x^2 - 16}$$

Answer: _____



Answer Key & Solutions

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TEACHER NOTES

#1: split fraction, integrate term by term. #5: long division first. #10: $u=\sqrt{x}$. #11: arctan; bounds $1/\sqrt{3} \rightarrow \pi/6$, $1 \rightarrow \pi/4$. #18: integrate $f'(x)$, use $f(1)=2$ to find $C=35/24$. #19: $f(x)=x^5-x^2+6$; $f(2)=32-4+6=34$. #21: num->12, den->0; DNE. #22-23: factor and cancel.

1 Evaluate: $\int_2^4 \frac{15x-3}{5x} dx$ $= 6 - \frac{3}{5} \ln 2$	2 Evaluate: $\int_0^1 x^6(4x^2 + x - 5) dx$ $= -\frac{73}{504}$	3 Evaluate: $\int 4 \sin^2 x dx$ $= 2x - \sin 2x + C$
4 Evaluate: $\int_3^4 \frac{3x}{x^2-2} dx$ $= \frac{3}{2} \ln 2$	5 Evaluate: $\int_1^x \frac{x^2}{1-x} dx$ $= -\frac{x^2}{2} - x - \ln x-1 + C$	6 Evaluate: $\int_{e^{5x}}^7 dx$ $= -\frac{7}{5} e^{-5x} + C$
7 Evaluate: $\int \cos^3(7x) dx$ $= \frac{1}{7} \sin 7x - \frac{1}{21} \sin^3 7x + C$	8 Evaluate: $\int \cos^5 x \sin x dx$ $= -\frac{1}{6} \cos^6 x + C$	9 Evaluate: $\int_{\ln 2}^{\ln 9} 4e^x dx$ $= 28$
10 Evaluate: $\int_4^{25} \frac{e^{\sqrt{x}}}{\sqrt{x}} dx$ $= 2(e^5 - e^2)$	11 Evaluate: $\int_{1/\sqrt{3}}^1 \frac{dx}{3+1+x^2}$ $= \frac{\pi}{12}$	12 Evaluate: $\int \cos(3-2x) dx$ $= -\frac{1}{2} \sin(3-2x) + C$
13 Evaluate: $\int \frac{\ln^2 x}{x} dx$ $= \frac{1}{3} (\ln x)^3 + C$	14 Evaluate: $\int \frac{3x}{(x^2+1)^2} dx$ $= -\frac{1}{2(x^2+1)} + C$	15 Evaluate: $\int \frac{3t-t^2}{2t} dt$ $= \frac{3}{2} t - \frac{1}{4} t^2 + C$
16 Evaluate: $\int \cos^3 2u \sin 2u du$ $= -\frac{1}{8} \cos^4 2u + C$	17 Find the antiderivative of: $\frac{2 \cos x}{\sin^2 x}$ $= -2 \csc x + C$	18 Given $f'(x)$ and $f(1)=2$, find $f(x)$: $f'(x) = \frac{1}{2}x^2 + \frac{3}{4}x$ $= \frac{x^3}{6} + \frac{3}{8}x^2 + \frac{35}{24}$
19 Given $f(x)$ and $f(1)=6$, find $f(2)$: $f'(x) = 5x^4 - 2x$ $= f(2) = 34$	20 Simplify: $e^{2 \ln x^3}$ $= x^6$	21 Evaluate the limit: $\lim_{x \rightarrow 4} \frac{x^2-4}{x^2-16}$ $= \text{DNE}$
22 Evaluate the limit: $\lim_{x \rightarrow 6} \frac{x-6}{x^2-36}$ $= \frac{1}{12}$	23 Evaluate the limit: $\lim_{x \rightarrow 4} \frac{x^3-64}{x^2-16}$ $= 6$	

